

School of Computer Science and Information Technology
Lucerne University of Applied Sciences and Arts (Switzerland)

Using Artificial Intelligence for Skin Cancer Diagnosis: Accuracy, Challenges, and Opportunities

By:
Lala Rud

Bachelor Thesis

Lucerne University of Applied Sciences and Arts
Bachelor of Science in International IT Management
Thesis Advisor: Dr. Zofia Baranczuk-Turska

24th May 2026

Declaration

Bachelor's Thesis at Lucerne University of Applied Sciences and the Arts School of Computer Science and Information Technology

Title of Bachelor's Thesis: Using Artificial Intelligence for Skin Cancer Diagnosis: Accuracy, Challenges, and Opportunities

Name of Student: Lala Rud

Degree Program: Bachelor's in International IT Management

Year of Graduation: 2026

Main Advisor: Prof. Dr. Zofia Barannzuk-Turska

External Expert: None

Industry partner/provider: None

Code/Thesis Classification

☐ Public (Standard)

☐ Private

Declaration

I hereby declare that I have completed this thesis alone and without any unauthorized or external help. I further declare that all the sources, references, literature, and any other associated resources have been correctly and appropriately cited and referenced. The confidentiality of the project provider (industry partner), as well as the intellectual property rights of the Lucerne University of Applied Sciences and the Arts have been fully and entirely respected in the completion of this thesis.

Place/Date, Signature_____

Expression of thanks and gratitude

I would like to express my sincere gratitude to my supervisor for the valuable guidance, support, and constructive feedback throughout the development of this bachelor's thesis. Your advice and encouragement helped me improve my work and complete this thesis successfully.

I would also like to thank my family and friends for their continuous support, patience, and motivation during my studies. Their encouragement gave me strength during challenging moments and helped me stay focused until this thesis was completed.

I am truly grateful to everyone who supported me throughout this journey.

Lala Rud, 2025

*Intellectual property of the degree programs of the Lucerne University of Applied
Sciences and Arts, FH Zentralschweiz, in accordance with Student Regulations:
Studienordnung*

Summary

This thesis investigates the use of deep learning-based image recognition for skin lesion classification using the publicly available HAM10000 dataset. The study evaluates how convolutional neural network architectures can classify dermoscopic images and how dataset imbalance affects model performance. The experiments focus on a reduced five-class classification task consisting of akiec, bcc, bkl, mel, and nv.

Two architectures, ResNet50 and EfficientNetB0, were compared under two capped class-weight configurations: 2.0 and 3.0. The models were evaluated using overall accuracy, balanced accuracy, macro F1-score, weighted F1-score, melanoma recall, melanoma specificity, and class-wise recall. These metrics were selected because the dataset is strongly imbalanced, and overall accuracy alone can hide poor performance on minority classes such as melanoma.

The final comparison showed that EfficientNetB0 with a capped class weight of 3.0 achieved the strongest overall trade-off among the tested configurations. It performed best on the main class-balanced and melanoma-focused metrics, including the highest melanoma recall. Although ResNet50 2.0 achieved the highest melanoma specificity, its melanoma recall was substantially lower. Therefore, EfficientNetB0 3.0 was selected as the best final model configuration for the screening-oriented objective of this thesis.

The results indicate that AI-based skin lesion classification can provide useful experimental insights, but the models remain below the diagnostic reliability expected in clinical practice. The findings highlight the importance of dataset quality, class balance, sensitivity-specificity trade-offs, and careful metric selection when evaluating AI systems for medical image analysis.

Summary	4
1. Introduction	6
2. Research Context, Literature Review, and Significance	7
2.1 Research context	7
2.2 Dermatological Accuracy	7
2.3 Dataset Limitations	8
2.4 Research Gap	8
3. Research Question	9
4. Methodology	10
4.1 Dataset and Preprocessing	10
4.2 Model Architectures	13
4.3 Experimental Design and Class Weighting	14
4.4 Model Training	15
4.4.1 Summary of Experimental Configurations	16
4.5 Evaluation Metrics	17
4.6 Summary of Experimental Design	18
5. Results	20
5.1 Preliminary Comparison: Seven-Class vs Five-Class ResNet	20
5.2 Final Five-Class Architecture Comparison	21
5.3 Class-Wise Recall Comparison	22
5.4 Effect of Class Weighting	23
5.5 Best Model Selection	24
6. Discussion	27
6.1 Impact of Class Imbalance on Model Performance	27
6.2 Sensitivity and Specificity Performance	28
6.3 Ethical Considerations	29
6.4 Interpretability and Clinical Trust	30
6.5 Summary of Findings	30
7. Conclusion	32
7.1 Future Work	33
8. Deliverables	34
Appendix A: Training Curves	36
Appendix B: Confusion Matrices	44

1. Introduction

Skin cancer is among the most frequently diagnosed cancers worldwide, and detecting it early plays a decisive role in improving a patient's chances of successful treatment. Countries with well-developed healthcare systems usually manage skin cancer effectively, yet the picture is very different in regions with scarce medical resources. In many low-income areas, people have limited access to dermatologists, diagnostic equipment, or regular screenings. As a result, cases that could have been identified earlier are often discovered only once the disease has advanced, which increases the risk of complications and raises overall mortality.

In Switzerland, by contrast, medical care is widely accessible and dermatological services are of high quality. Even so, early detection still relies on individuals choosing to seek out examinations. Many people do not prioritize routine skin checks and consult a professional only after noticing visible changes. This behavior can unintentionally delay the identification of lesions that might require closer attention. Because of this, there is growing interest in digital tools that could help people examine their skin more proactively, especially in situations where a simple initial assessment might encourage them to contact a specialist sooner.

At the same time, ideas such as fully automated, at-home diagnosis remain far from practical use. Introducing such systems would require meeting strict medical, ethical, and regulatory standards and demonstrating a very high degree of reliability. For these reasons, the aim of this thesis will not be to propose a replacement for dermatologists, but rather to explore how image-based machine learning models might support the diagnostic process. The focus is on evaluating how accurately such models can classify skin lesions and identifying the factors that influence their performance. A model that behaves consistently and has apparent limitations could eventually contribute to the development of tools designed to assist, rather than replace, medical professionals.

2. Research Context, Literature Review, and Significance

2.1 Research context

In recent years, AI models have achieved high accuracy in medical image analysis, including melanoma detection (Esteva et al., 2017; Haenssle et al., 2018). However, many of these systems still operate as “black boxes”, making it difficult for clinicians to understand how decisions are reached. This lack of transparency creates hesitation in real clinical use, especially when decisions involve patient safety and legal responsibility. For these reasons, researchers and clinicians increasingly argue that diagnostic support tools must offer explanations that are understandable and relevant to medical practice.

A recent study by Chanda et al. (2024) clearly illustrates this shift. The authors developed an explainable AI (XAI) model that bases its decisions on well-known dermoscopic criteria and shows clinicians which features influenced the prediction. In a large three-phase study with 116 dermatologists, the system did not raise diagnostic accuracy beyond standard AI assistance, but it did have an important effect: clinicians reported substantially higher confidence and trust when explanations were provided. This result indicates that interpretability is not only an ethical or regulatory requirement, but also a practical factor in whether professionals actually accept and use AI tools. As a result, explainability has become a central topic in current research on the safe integration of AI into high-stakes medical environments.

In this thesis, interpretability is not implemented as a separate experimental component but is considered an important limitation and a future requirement when discussing the clinical relevance of the model results.

2.2 Dermatological Accuracy

Several studies have reported dermatologists' diagnostic accuracy in evaluating skin lesions, providing useful reference points for indirect comparisons. Haenssle et al. (2018) found that experienced dermatologists achieved sensitivity levels of 86–88% for melanoma detection when using dermoscopic images, while specificity ranged from 71–74%. Other studies report broader ranges depending on clinical experience, image type, and available context, with sensitivity varying between 70–90% across practitioners (Vestergaard et al., 2008; Kittler et al., 2002). These results highlight that expert performance is strong but not uniform, and that both false positives and false negatives remain common in real

diagnostic settings. Because of this variability, indirect comparisons between AI performance and dermatologists' ranges are meaningful, even without direct access to clinical evaluation data

2.3 Dataset Limitations

Public dermatology datasets such as HAM10000 are widely used in academic research, but they come with several limitations that affect how AI models generalize (Tschandl et al., 2018). The dataset is heavily imbalanced, with some lesion types appearing far more frequently than others, potentially biasing model predictions toward the majority classes. In addition, the images are primarily from fair-skinned patients (Fitzpatrick types I–III), creating limited representation of darker skin tones and potentially reducing performance for underrepresented populations (Daneshjou et al., 2022). The dataset also consists almost entirely of high-quality dermoscopic images rather than smartphone photographs, meaning the visual conditions are cleaner and more consistent than those in real patient environments. These factors should be considered when evaluating the accuracy of any model trained on HAM10000.

2.4 Research Gap

Although many studies demonstrate strong AI performance for skin lesion classification, most evaluations rely on curated datasets and controlled imaging conditions (Esteva et al., 2017; Tschandl et al., 2019). At the same time, explainability, dataset bias, and clinical variability continue to limit adoption in real medical practice (Brancaccio et al., 2024). There is therefore a need for work that examines how modern deep learning models perform on publicly available datasets, reflects on their strengths and weaknesses, and situates their results within the broader challenges highlighted in current research. This thesis contributes to that gap by training and evaluating convolutional neural networks on the HAM10000 dataset and discussing how accuracy, limitations, and dataset constraints relate to the opportunities and challenges of using AI in skin cancer diagnosis.

3. Research Question

How accurately can deep learning-based image recognition models classify skin lesion images when trained on the publicly available HAM10000 dataset, and how does their performance compare with the reported diagnostic accuracy ranges of dermatologists?

This question evaluates whether convolutional neural network architectures can produce meaningful classification results for skin lesion diagnosis under academic experimental conditions. The comparison is not intended to claim clinical equivalence with dermatologists, since the models are trained and tested on a curated public dataset rather than evaluated in a real clinical workflow. Instead, dermatologist performance reported in the literature is used as an indirect reference point for interpreting the model results. The comparison is therefore interpretive rather than clinical: the models are not evaluated against dermatologists on the same cases, and the results should not be understood as evidence of clinical equivalence.

To support the main research question, the thesis further investigates the following sub-questions:

- **RQ1.1:** How do dataset characteristics, especially class imbalance, skin-type representation, and dermoscopic image quality, affect the training and evaluation of AI-based skin lesion classification models?
- **RQ1.2:** How do the two selected deep learning architectures, ResNet and EfficientNet, perform relative to each other on the same reduced HAM10000 classification task?
- **RQ1.3:** How does the use of different class-weight settings influence the comparison between ResNet and EfficientNet, particularly for minority classes such as melanoma?
- **RQ1.4:** Which model configuration provides the most suitable trade-off between overall accuracy, balanced accuracy, melanoma sensitivity, and class-wise performance?

The class-weight comparison is treated as an experimental dimension within the architecture comparison. Since the HAM10000 dataset is imbalanced, class weighting is used to reduce the tendency of models to favor the dominant class. However, the experiments do not assume that the same weighting intensity is

equally suitable for all architectures. Instead, the effect of weighting is interpreted relative to each model's actual classification behavior, particularly its ability to detect minority classes such as melanoma.

4. Methodology

This chapter outlines the research design and systematic approach used to develop, train, and evaluate the deep learning models applied in this thesis. The study will follow an experimental, quantitative research methodology commonly used in computer vision and medical image analysis. This approach will enable objective measurement of model effectiveness, comparison of different architectures, and analysis of how dataset attributes influence final results. The aim is to develop supervised models and examine their capabilities and limitations using publicly available data.

This chapter describes the experimental technique used to train, evaluate, and compare deep learning models for skin lesion classification. The study follows a quantitative experimental design in which two convolutional neural network architectures, ResNet and EfficientNet, are trained and evaluated on a reduced version of the HAM10000 dataset.

The main purpose of the methodology is to compare how the two architectures perform under the same classification task and evaluation framework. Since the dataset is strongly imbalanced, class weighting is included as an additional experimental dimension. This allows the study to examine not only which architecture performs better overall but also how each architecture responds to imbalance handling during training.

The evaluation focuses on overall accuracy, balanced accuracy, macro F1-score, weighted F1-score, sensitivity, specificity, and class-wise recall. These metrics are necessary because medical image classification cannot be evaluated reliably using accuracy alone. In imbalanced datasets, a model may achieve acceptable accuracy by mainly predicting the dominant class while still failing to detect clinically important minority classes such as melanoma.

4.1 Dataset and Preprocessing

The experiments presented in this thesis utilize the HAM10000 dataset, a publicly available benchmark for skin lesion classification made available through Kaggle (Mader, n.d.). HAM10000 comprises 10,015 dermoscopic images of pigmented skin lesions and is widely used in academic machine learning research focused on dermatological image classification.

The original dataset includes seven diagnostic categories: actinic keratoses and intraepithelial carcinoma (akiec), basal cell carcinoma (bcc), benign keratosis-like lesions (bkl), dermatofibroma (df), melanoma (mel), melanocytic nevi (nv), and vascular lesions (vasc). For the final architecture comparison, a reduced five-class setup was employed, consisting of akiec, bcc, bkl, mel, and nv.

Figure 1 shows the class distribution of the original seven-class HAM10000 dataset. The distribution is strongly imbalanced, with the nv class containing far more images than all other categories. In contrast, vasc and df contain very few samples. This imbalance is important because it can cause a model to prioritize the majority class while learning weak representations for minority classes.

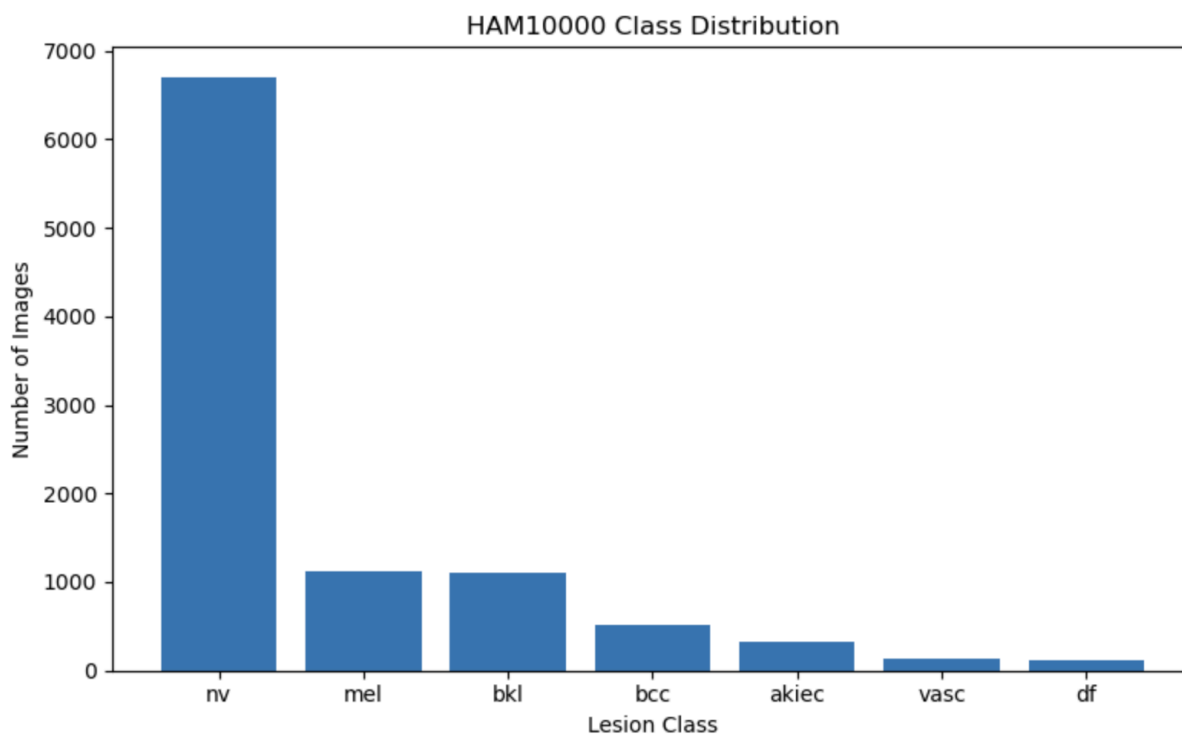


Figure 1. Class distribution of the original seven-class HAM10000 dataset.

The reduced five-class setup was chosen due to the severe underrepresentation of the df and vasc classes, which resulted in unstable training behavior during preliminary experiments. Although removing these two classes did not fully resolve the class imbalance, it mitigated the most significant instability and enabled a more meaningful comparison between ResNet and EfficientNet. The classification task remains both clinically and technically challenging, as the dataset continues to exhibit a pronounced imbalance between the dominant nv class and minority classes such as melanoma, bcc, and akiec.

Figure 2 shows the reduced five-class dataset used for the final architecture comparison. The df and vasc classes were removed because they were severely underrepresented and contributed to unstable training

behavior in preliminary experiments. The reduced dataset remains imbalanced, especially because the nv class is still much larger than the other classes. Therefore, class weighting and class-wise evaluation metrics remain necessary.

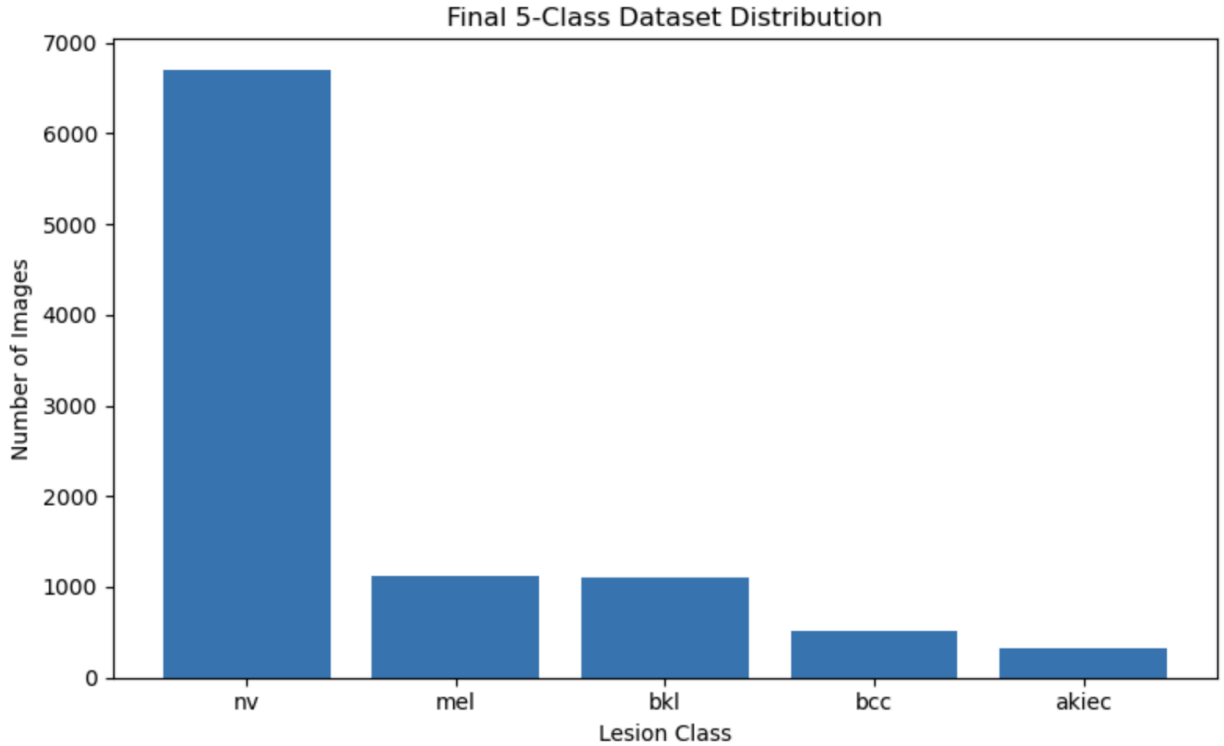


Figure 2. Class distribution of the reduced five-class dataset used for the final ResNet and EfficientNet comparison.

The comparison between the two distributions explains why the final experiments used a five-class setup. Removing df and vasc reduced the most extreme class imbalance, but it did not eliminate the imbalance completely. The final task therefore, remained suitable for evaluating how ResNet and EfficientNet handle minority-class detection under imbalanced conditions.

Prior to training, all images were resized to match the input dimensions required by the selected convolutional neural networks. Pixel values were normalized to ensure compatibility with pretrained ImageNet weights. Data augmentation techniques were applied to enhance generalization and reduce overfitting. The dataset was divided into training, validation, and test sets using a stratified approach to preserve class distribution across the splits as much as possible.

The test set was maintained separately from the training and validation processes and was used exclusively for final model evaluation. This approach ensured that the reported performance metrics

reflect the model's ability to generalize to previously unseen images rather than its capacity to memorize training examples.

This chapter outlines the experimental methodology employed to train, evaluate, and compare deep learning models for skin lesion classification. The study adopts a quantitative experimental design in which two convolutional neural network architectures, ResNet and EfficientNet, are trained and evaluated on a reduced version of the HAM10000 dataset.

4.2 Model Architectures

Two convolutional neural network architectures were selected for comparison: ResNet and EfficientNet. Both architectures are commonly used for image classification and are suitable for transfer learning, but they are based on different design principles.

A convolutional neural network, or CNN, is a neural network designed for image data. Instead of treating an image only as a flat list of pixels, convolutional layers scan small regions of the image and learn filters that detect visual patterns. Early layers usually detect simple structures such as edges, color changes, and textures, while deeper layers combine these patterns into more complex image features. This is relevant for skin lesion classification because the model must learn visual differences between lesion categories, such as color variation, texture, shape, and border irregularity.

ResNet was selected as the baseline architecture because it is a well-established CNN model and introduced the idea of residual learning. In very deep networks, training can become difficult because information and gradients must pass through many layers. ResNet addresses this problem by using residual connections, also called skip connections, which allow information to bypass some layers and move more directly through the network. He et al. describe residual learning as a method that makes very deep networks easier to optimize and allows them to benefit from increased depth. This makes ResNet a suitable baseline for evaluating whether a widely used CNN architecture can classify skin lesion images under imbalanced dataset conditions (He et al., 2015).

EfficientNet was selected as the second architecture because it follows a different scaling strategy. Tan and Le argue that CNN performance can be improved by balancing three dimensions of the model: depth, width, and image resolution. Depth refers to the number of layers, width refers to the number of channels or filters, and resolution refers to the input image size. Instead of increasing only one of these dimensions, EfficientNet uses compound scaling to scale them together in a structured way. This design aims to improve the balance between accuracy and computational efficiency (Tan & Le, 2019).

The comparison between ResNet and EfficientNet therefore compares two different approaches to image feature learning. ResNet represents a strong residual-learning baseline, while EfficientNet represents a more recent architecture based on balanced model scaling. This comparison is relevant for the thesis because architecture choice may influence how well a model learns minority-class patterns in an imbalanced medical dataset.

4.3 Experimental Design and Class Weighting

The experimental design was developed in stages. The first experiments used ResNet on the original seven-class HAM10000 dataset, which includes akiec, bcc, bkl, df, mel, nv, and vasc. This stage was used to evaluate how the model behaved under the full dataset structure and to identify the main difficulties caused by class imbalance.

The seven-class experiments showed that the most underrepresented classes, especially df and vasc, contributed to unstable training behavior and weak minority-class detection. For this reason, the classification task was reduced to five classes by removing df and vasc. The reduced five-class setup retained akiec, bcc, bkl, mel, and nv. This allowed the experiments to focus on clinically relevant and better-represented classes while still preserving the central imbalance problem caused by the dominant nv class.

After the five-class setup was established, ResNet was trained again using the reduced dataset. From this point onward, the final architecture comparison was conducted only on the five-class task. EfficientNet was then introduced as a second architecture and compared with ResNet under the same five-class classification setup.

Class weighting was included as an additional experimental dimension in the final five-class comparison. Class weighting changes how strongly errors from each class contribute to the training loss. Since nv was the dominant class, a model could otherwise reduce its loss mainly by learning to classify nv correctly while still performing poorly on minority classes. To reduce this effect, higher weights were assigned to minority classes and a lower weight was assigned to nv.

Two capped class-weight configurations were tested: 2.0 and 3.0. The cap defines the maximum weight assigned to the smallest minority classes. In the 2.0 configuration, the maximum class weight was limited to 2.0. In the 3.0 configuration, the maximum class weight was increased to 3.0. This affected the smallest minority classes, akiec and bcc, while the weights for bkl, mel, and nv remained unchanged because their calculated weights were already below the cap.

Class	Label Index	Weight 2.0 Configuration	Weight 3.0 Configuration
akiec	0	2.0000	3.0000
bcc	1	2.0000	3.0000
bkl	2	1.7762	1.7762
mel	3	1.7541	1.7541
nv	4	0.2911	0.2911

Table 1. Capped class weights used in the final five-class experiments

Using these two configurations resulted in four final five-class model configurations: ResNet with capped class weight 2.0, ResNet with capped class weight 3.0, EfficientNet with capped class weight 2.0, and EfficientNet with capped class weight 3.0.

Testing both weighting configurations made it possible to observe whether stronger weighting improved minority-class detection and whether ResNet and EfficientNet responded differently to the same imbalance-handling strategy. Therefore, the main comparison in the thesis remains between ResNet and EfficientNet, while the class-weight settings serve as a supporting dimension to better understand each architecture’s behavior under imbalanced conditions.

4.4 Model Training

Both ResNet and EfficientNet were trained using transfer learning with pretrained ImageNet weights. Transfer learning was selected because the HAM10000 dataset is relatively small compared to large-scale general image datasets, and pretrained convolutional layers can provide useful low-level visual feature extraction.

For each architecture, the classification head was adapted to the five-class skin lesion task. The final output layer used five neurons corresponding to the selected classes: akiec, bcc, bkl, mel, and nv. The models were trained using the training set, monitored on the validation set, and finally evaluated on the unseen test set.

The experimental process was conducted in two stages. First, exploratory experiments were performed using the original seven-class structure of the HAM10000 dataset. These experiments were important because they showed how strongly the extremely underrepresented classes affected model stability and minority-class detection. In particular, dermatofibroma (df) and vascular lesions (vasc) were underrepresented, which contributed to unstable training behavior and poor class-wise recall.

Class weighting was implemented during training to mitigate the influence of the majority class. Two weighting intensities, 2.0 and 3.0, were tested for each architecture to assess whether increased weighting enhanced minority-class detection and to compare the effects between ResNet and EfficientNet.

The models were evaluated using confusion matrices and classification reports. Overall accuracy was included, but it was not treated as the main performance indicator because the dataset is imbalanced. Instead, the evaluation focused on balanced accuracy, macro F1-score, weighted F1-score, sensitivity, specificity, and class-wise recall. Particular attention was given to melanoma recall, since missing melanoma cases pose a greater risk in a screening context than incorrectly flagging benign cases for further examination.

Earlier experiments using the full seven-class setup showed unstable behavior, especially for extremely underrepresented classes. These experiments informed the final five-class setup and confirmed the importance of class-wise evaluation. The final comparison, therefore, focuses on the four controlled five-class experiments: ResNet 2.0, ResNet 3.0, EfficientNet 2.0, and EfficientNet 3.0.

In earlier seven-class experiments, extremely underrepresented classes, such as dermatofibroma (df) and vascular lesions (vasc), achieved very poor recall values, contributing to unstable model behavior. This highlights a key limitation of deep learning approaches when applied to highly imbalanced datasets.

The confusion matrices across multiple experiments confirmed that class imbalance strongly affected predictive performance, often leading the model to favor dominant lesion categories, such as nevi (nv).

4.4.1 Summary of Experimental Configurations

The experimental process included preliminary seven-class experiments and a final controlled five-class architecture comparison. The preliminary experiments were used to identify instability caused by extremely underrepresented classes, especially df and vasc. The final comparison focused on the reduced five-class dataset and evaluated ResNet50 and EfficientNetB0 under two capped class-weight configurations.

Experiment type	Model	Classes	Class-weight cap	Purpose
Preliminary setup	ResNet50	7	3.0	Evaluate the original HAM10000 class structure and identify instability caused by highly underrepresented classes.
Final comparison	ResNet50	5	2.0	Test ResNet50 under moderate class weighting.
Final comparison	ResNet50	5	3.0	Test ResNet50 under stronger class weighting.
Final comparison	EfficientNetB0	5	2.0	Test EfficientNetB0 under moderate class weighting.
Final comparison	EfficientNetB0	5	3.0	Test EfficientNetB0 under stronger class weighting.

Table 2. Final experimental configurations used for model comparison

The preliminary seven-class experiments informed the final five-class setup but were not used for final model selection. Final model selection was based on the four controlled five-class experiments: ResNet50 2.0, ResNet50 3.0, EfficientNetB0 2.0, and EfficientNetB0 3.0. This ensured that the final comparison evaluated both the choice of architecture and the effect of capped class weighting on the same reduced classification task.

4.5 Evaluation Metrics

Several metrics were used to evaluate model performance. Overall accuracy measures the proportion of images correctly classified across the entire test set. However, because the dataset is imbalanced, accuracy alone can be misleading. A model can achieve relatively high accuracy by correctly classifying the dominant nv class while performing poorly on minority classes.

Balanced accuracy was therefore used to measure the average recall across all classes. This gives each class equal importance, regardless of how many samples it contains. Macro F1-score was also used because it calculates the average F1-score across classes without weighting by class frequency. This makes it useful for evaluating whether the model performs consistently across minority and majority classes.

Weighted F1-score was included to show the overall F1 performance while accounting for class support. This metric is useful, but it can still be influenced by the dominant class. For this reason, it was interpreted together with macro F1-score and class-wise recall.

Sensitivity, also referred to as recall, was especially important for melanoma. Melanoma sensitivity measures the proportion of true melanoma cases that the model correctly identifies. Specificity was also calculated to evaluate how well the model avoided incorrectly classifying non-melanoma cases as melanoma.

The confusion matrix was used to analyze the error types made by each model. This made it possible to identify whether a model confused melanoma with benign keratosis-like lesions, nevi, or other classes. Such error patterns are important because models with similar accuracy can behave very differently from a medical perspective.

4.6 Summary of Experimental Design

The experimental process consisted of two main stages. The first stage used the original seven-class HAM10000 setup to evaluate the model's behavior on the full dataset. These experiments showed that the most underrepresented classes, especially df and vasc, contributed to unstable training behavior and weak minority-class detection.

The second stage used a reduced five-class setup for the final architecture comparison. This setup retained akiec, bcc, bkl, mel, and nv. It allowed the comparison to remain clinically relevant while reducing the instability caused by extremely small classes.

The final comparison focused on ResNet50 and EfficientNetB0 under two capped class-weight settings. This produced four final experimental configurations.

Stage	Model / Setup	Classes	Class Weight	Purpose

Preliminary	ResNet, original setup	7	3.0	Evaluate full HAM10000 class structure and identify instability
Final comparison	ResNet	5	2.0	Test ResNet under moderate weighting
Final comparison	ResNet	5	3.0	Test ResNet under stronger weighting
Final comparison	EfficientNet	5	2.0	Test EfficientNet under moderate weighting
Final comparison	EfficientNet	5	3.0	Test EfficientNet under stronger weighting

Table 3. Summary of the experimental design

This design supports two levels of analysis. First, the seven-class and five-class ResNet experiments show how dataset structure affects model behavior. Second, the five-class ResNet and EfficientNet experiments enable a direct architectural comparison on the same reduced classification task.

5. Results

This chapter presents the results of the experimental model comparison. The results are organized in two stages. First, the preliminary seven-class ResNet experiment is compared with the reduced five-class ResNet experiment in order to evaluate the effect of dataset structure. Second, the final five-class comparison between ResNet and EfficientNet is presented using class weights of 2.0 and 3.0.

The evaluation focuses on overall accuracy, balanced accuracy, macro F1-score, weighted F1-score, sensitivity, specificity, and class-wise recall. Particular attention is given to melanoma recall because melanoma detection is the most clinically relevant outcome in the context of skin cancer screening.

5.1 Preliminary Comparison: Seven-Class vs Five-Class ResNet

The first important comparison was between the original seven-class ResNet model and the reduced five-class ResNet model. Both experiments used a class weight of 3.0. This comparison was used to evaluate whether reducing the dataset from seven classes to five classes improved model stability and melanoma detection.

The seven-class setup followed the original HAM10000 class structure, including akiec, bcc, bkl, df, mel, nv, and vasc. However, the df and vasc classes were highly underrepresented. During training, this contributed to unstable validation behavior and weak recall for several minority classes. The reduced five-class setup removed df and vasc while retaining akiec, bcc, bkl, mel, and nv.

The comparison shows that reducing the task from seven classes to five classes improved the clarity of the evaluation, but it did not fully solve the minority-class problem for ResNet50. In the final five-class comparison, ResNet50 with capped class weight 3.0 achieved an overall accuracy of 0.648 and a melanoma recall of 0.435. Although this was an improvement over ResNet50 2.0, the balanced accuracy remained low at 0.391, and the macro F1-score was only 0.363. This indicates that ResNet50 still struggled to achieve consistent performance across classes, especially when compared with EfficientNetB0.

This result is important because it shows that the dataset structure was only one part of the problem. Removing df and vasc reduced the most extreme class imbalance, but the remaining five-class task was still strongly imbalanced because the nv class remained dominant. The latter comparison with EfficientNetB0 showed that architecture choice also influenced minority-class performance, since EfficientNetB0 achieved substantially higher balanced accuracy, macro F1-score, and melanoma recall than ResNet50.

5.2 Final Five-Class Architecture Comparison

The final model comparison used the reduced five-class setup with the class order akiec, bcc, bkl, mel, and nv. Both ResNet50 and EfficientNetB0 were evaluated using capped class-weight configurations of 2.0 and 3.0. This allowed comparison of both architecture performance and the effect of class weighting.

Model configuration	Accuracy	Balanced accuracy	Macro F1	Weighted F1	Melanoma recall	Melanoma Specificity
ResNet50 2.0	0.594	0.356	0.298	0.606	0.332	0.902
ResNet50 3.0	0.648	0.391	0.363	0.631	0.435	0.856
EfficientNetB0 2.0	0.646	0.516	0.474	0.679	0.517	0.837
EfficientNetB0 3.0	0.649	0.518	0.483	0.683	0.596	0.800

Table 4. Final five-class model comparison

The final comparison shows that EfficientNetB0 outperformed ResNet50 on the most relevant class-balanced and melanoma-focused metrics. Although ResNet50 3.0 achieved an overall accuracy of 0.648, which was close to the EfficientNetB0 results, its balanced accuracy, macro F1-score, and melanoma recall remained substantially lower. This confirms that raw accuracy alone was insufficient for evaluating model quality on the imbalanced five-class dataset.

EfficientNetB0 3.0 achieved the highest overall accuracy, balanced accuracy, macro F1-score, weighted F1-score, and melanoma recall. EfficientNetB0 2.0 also performed clearly better than both ResNet50 configurations on class-balanced metrics, but it achieved lower melanoma recall than EfficientNetB0 3.0.

The comparison demonstrates that similar accuracy values can hide very different classification behavior. ResNet50 3.0 and EfficientNetB0 3.0 achieved similar overall accuracy values, but EfficientNetB0 3.0 was much stronger in balanced accuracy and melanoma recall. This means that EfficientNetB0 classified minority classes more effectively and was less dependent on the correct classification of the dominant nv class. ResNet50 2.0 achieved the highest melanoma specificity, but its low melanoma recall indicates that

it missed many true melanoma cases. Therefore, specificity alone was not sufficient for selecting the final model.

Overall, EfficientNetB0 3.0 provided the strongest performance across the most important evaluation criteria. It was therefore selected as the best final model configuration for the screening-oriented objective of this thesis.

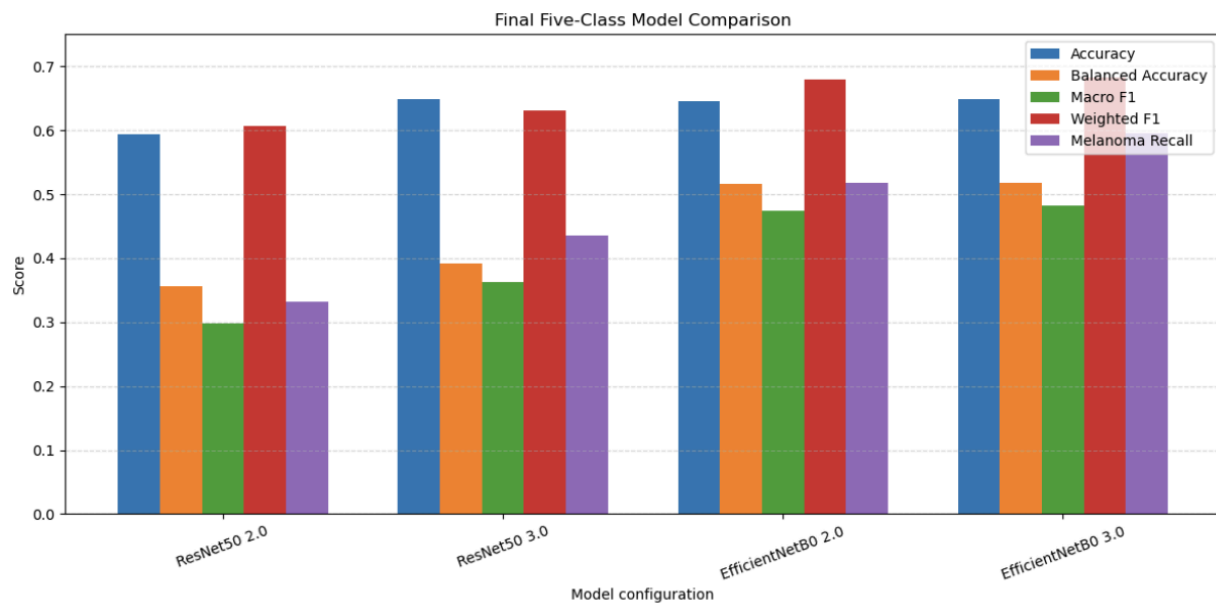


Figure 3. Comparison of final five-class model configurations across accuracy, balanced accuracy, macro F1-score, weighted F1-score, and melanoma recall.

5.3 Class-Wise Recall Comparison

Class-wise recall provides a clearer view of how each model performed across the individual diagnostic categories. This is especially important because the dataset is imbalanced and raw accuracy can hide weak minority-class performance.

Class	ResNet50 2.0	ResNet50 3.0	EfficientNetB0 2.0	EfficientNetB0 3.0
akiec	0.538	0.246	0.231	0.277
bcc	0.058	0.417	0.476	0.437

bkl	0.086	0.036	0.659	0.573
mel	0.332	0.435	0.517	0.596
nv	0.764	0.820	0.699	0.705

Table 5. Class-wise recall comparison for ResNet50 and EfficientNetB0 models

The class-wise recall comparison shows that the four configurations behaved differently across lesion classes. ResNet50 2.0 achieved the highest recall for akiec, but performed very weakly on bcc, bkl, and mel. ResNet50 3.0 improved melanoma recall compared with ResNet50 2.0, increasing from 0.332 to 0.435, but it still remained below both EfficientNetB0 configurations.

EfficientNetB0 showed stronger minority-class performance overall. EfficientNetB0 2.0 achieved a strong recall for bcc and bkl, while EfficientNetB0 3.0 achieved the highest melanoma recall at 0.596. This result is especially important because melanoma detection is the most clinically relevant outcome in this thesis. Although EfficientNetB0 3.0 did not achieve the highest recall for every class, it delivered the strongest melanoma-focused result and achieved the best overall combination of balanced accuracy, macro F1-score, weighted F1-score, and test accuracy.

5.4 Effect of Class Weighting

The effect of increasing the capped class weight from 2.0 to 3.0 differed between ResNet50 and EfficientNetB0, but in both architectures, the stronger weighting improved melanoma recall. This indicates that the higher class-weight cap made the models more sensitive to minority-class cases, especially melanoma. However, this improvement came with a reduction in melanoma specificity, showing a trade-off between detecting more true melanoma cases and incorrectly classifying more non-melanoma lesions as melanoma.

For ResNet50, increasing the capped class weight from 2.0 to 3.0 improved several important metrics. Overall accuracy increased from 0.594 to 0.648, balanced accuracy increased from 0.356 to 0.391, macro F1-score increased from 0.298 to 0.363, and melanoma recall increased from 0.332 to 0.435. This shows that stronger weighting helped ResNet50 detect more melanoma cases and improved its average class-wise performance. However, melanoma specificity decreased from 0.902 to 0.856, indicating that the model produced more false-positive melanoma predictions after the stronger weighting was applied.

For EfficientNetB0, increasing the capped class weight from 2.0 to 3.0 also improved most key evaluation metrics. Accuracy increased from 0.646 to 0.649, balanced accuracy increased from 0.516 to 0.518, macro F1-score increased from 0.474 to 0.483, weighted F1-score increased from 0.679 to 0.683, and melanoma recall increased from 0.517 to 0.596. This means that EfficientNetB0 3.0 detected a larger proportion of melanoma cases than EfficientNetB0 2.0. However, melanoma specificity decreased from 0.837 to 0.800, again showing that higher sensitivity came at the cost of more false positives.

These results show that class weighting should not be interpreted only as a simple correction for dataset imbalance. Instead, it should be understood as a model-dependent training intervention that changes the balance between sensitivity and specificity. In this thesis, stronger weighting improved melanoma recall for both architectures, but it also reduced melanoma specificity. Since the thesis has a screening-oriented focus, the increase in melanoma recall is particularly significant, because missing melanoma cases is more clinically serious than incorrectly flagging benign lesions for further examination.

5.5 Best Model Selection

Based on the final five-class comparison, **EfficientNetB0 with a capped class weight 3.0** was selected as the best final model configuration in this thesis. The selection was not based on accuracy alone, because the HAM10000 dataset is imbalanced and high accuracy can be strongly influenced by the correct classification of the dominant nv class. Instead, the decision was based on the combined interpretation of overall accuracy, balanced accuracy, macro F1-score, weighted F1-score, melanoma recall, melanoma specificity, and class-wise performance.

Metric	Value
Test accuracy	0.649
Balanced accuracy	0.518
Macro precision	0.486
Macro recall	0.518
Macro F1-score	0.483
Weighted precision	0.758

Weighted recall	0.649
Weighted F1-score	0.683
Melanoma precision	0.278
Melanoma recall / sensitivity	0.596
Melanoma specificity	0.800
Melanoma F1-score	0.379

Table 6. Detailed final metrics for the selected EfficientNetB0 3.0 model

EfficientNetB0 3.0 achieved the strongest overall result among the four final configurations. It achieved the highest test accuracy (0.649), balanced accuracy (0.518), macro F1-score (0.483), weighted F1-score (0.683), and melanoma recall (0.596). These results indicate that EfficientNetB0 3.0 was not only slightly stronger in overall classification performance, but also more effective at detecting melanoma cases than the other tested configurations.

The selection of EfficientNetB0 3.0 is especially relevant for the screening-oriented objective of this thesis. In a medical screening context, missing melanoma cases is more critical than incorrectly flagging benign lesions for further examination. Although ResNet50 2.0 achieved the highest melanoma specificity at 0.902, its melanoma recall was only 0.332. This distinction is important because a model with high specificity can still be unsuitable for screening if it fails to detect a sufficient proportion of true melanoma cases. This means that ResNet50 2.0 was better at avoiding false melanoma predictions, but it missed a larger proportion of true melanoma cases.

EfficientNetB0 3.0, by contrast, achieved a stronger melanoma recall of 0.596 while maintaining a specificity of 0.800. Therefore, EfficientNetB0 3.0 provided a more suitable balance for the diagnostic-support scenario considered in this thesis.

This result also answers the four supporting research questions. For RQ1.1, the results confirm that class imbalance strongly affected model behavior, because high overall accuracy did not always correspond to strong minority-class performance. For RQ1.2, EfficientNetB0 outperformed ResNet50 on the reduced five-class HAM10000 task, particularly in balanced accuracy, macro F1-score, and melanoma recall. For RQ1.3, increasing the capped class weight from 2.0 to 3.0 improved melanoma recall for both ResNet50

and EfficientNetB0, but also reduced melanoma specificity. This confirms that class weighting influenced the sensitivity–specificity trade-off. For RQ1.4, EfficientNetB0 3.0 provided the most suitable trade-off between overall accuracy, balanced accuracy, melanoma sensitivity, and class-wise performance.

Evaluation criterion	Best model configuration	Interpretation
Highest overall accuracy	EfficientNetB0 3.0	Best general test-set accuracy
Highest balanced accuracy	EfficientNetB0 3.0	Best average recall across classes
Highest macro F1-score	EfficientNetB0 3.0	Best average F1-score across classes
Highest weighted F1-score	EfficientNetB0 3.0	Best support-weighted F1-score
Highest melanoma recall	EfficientNetB0 3.0	Best melanoma detection
Highest melanoma specificity	ResNet50 2.0	Best at avoiding false melanoma predictions
Best ResNet configuration	ResNet50 3.0	Stronger ResNet trade-off than ResNet50 2.0
Selected final model	EfficientNetB0 3.0	Best screening-oriented trade-off

Table 7. Best-performing model by evaluation criterion

6. Discussion

This section interprets the experimental results in relation to the research question and existing literature. The objective is not only to report performance metrics, but to explain what they imply about the capabilities and limitations of AI-based skin cancer classification.

6.1 Impact of Class Imbalance on Model Performance

One of the most significant challenges encountered in this research was the severe class imbalance in the HAM10000 dataset. Certain lesion categories, particularly benign nevi (nv), contained substantially more images than minority classes such as dermatofibroma (df) and vascular lesions (vasc). This imbalance strongly influenced the behavior of the deep learning models throughout the experiments.

In the initial baseline experiments, the ResNet50 model achieved deceptively high overall accuracy while failing to correctly classify minority classes. In some configurations, the model predicted that almost all samples belonged to the dominant nevi (nv) class, resulting in melanoma recall values near 0%. This demonstrated that overall accuracy alone was insufficient for evaluating medical classification systems trained on imbalanced datasets.

To reduce this bias, class weighting techniques were introduced during training. Although class weighting improved the importance assigned to minority classes, it also introduced instability into the learning process. In several experiments, the model began to overpredict underrepresented classes, causing large fluctuations in validation accuracy and inconsistent classification behavior.

The experiments suggest that class imbalance should not be treated only as a dataset problem, but also as a model-dependent problem. Different architectures may respond differently to the same imbalanced dataset and the same class-weight configuration. In the final comparison, increasing the capped class weight from 2.0 to 3.0 improved melanoma recall for both ResNet50 and EfficientNetB0, but it also reduced melanoma specificity. This shows that class weighting influenced the sensitivity-specificity balance rather than simply improving all aspects of performance.

This means that class weights should be selected based on each model's actual behavior, especially class-wise recall, melanoma sensitivity, melanoma specificity, and confusion matrix patterns. Weighting is not only a correction for class distribution; it is also a training intervention that changes how the model balances false negatives and false positives.

The experiments also demonstrated that extremely underrepresented classes negatively affected model stability and generalization. Removing dermatofibroma (df) and vascular lesions (vasc) reduced the most extreme imbalance and made the final comparison more meaningful, but it did not eliminate the imbalance problem. In the final five-class setup, performance still varied strongly between classes and architectures. Melanoma recall ranged from 0.332 in ResNet50 2.0 to 0.596 in EfficientNetB0 3.0. This shows that minority-class performance depended not only on the dataset structure but also on the choice of architecture and the class-weight configuration.

These findings confirm observations from existing literature that dataset imbalance remains one of the primary limitations of medical deep learning systems. Even when transfer learning and class-balancing techniques are applied, minority-class representation still significantly affects model reliability and diagnostic consistency.

6.2 Sensitivity and Specificity Performance

A central finding of this research is the trade-off between sensitivity and specificity in AI-based skin lesion classification. In medical diagnosis, sensitivity measures the proportion of positive cases correctly identified by the model, while specificity measures the proportion of negative cases correctly identified as negative. Both metrics are important for melanoma detection because false negatives and false positives have different clinical consequences.

Across the final five-class experiments, melanoma recall varied substantially between model configurations. ResNet50 2.0 achieved a melanoma recall of 0.332, while ResNet50 3.0 improved this value to 0.435. EfficientNetB0 performed better on melanoma detection, with EfficientNetB0 2.0 achieving a melanoma recall of 0.517 and EfficientNetB0 3.0 achieving the highest melanoma recall of 0.596. This means that the final selected model, EfficientNetB0 3.0, correctly identified approximately 59.6% of melanoma cases in the test set.

However, the strongest melanoma recall did not correspond to the highest melanoma specificity. ResNet50 2.0 achieved the highest melanoma specificity of 0.902, indicating it was best at avoiding incorrect melanoma predictions among non-melanoma cases. However, its melanoma recall was only 0.332, meaning that it missed a large proportion of true melanoma cases. EfficientNetB0 3.0 achieved lower melanoma specificity (0.800) but substantially higher melanoma recall (0.596). This illustrates the central sensitivity-specificity trade-off in the experiments.

For a screening-oriented system, the EfficientNetB0 3.0 trade-off is more appropriate than the ResNet50 2.0 trade-off. In screening, missing melanoma cases is more serious than incorrectly referring benign lesions for further examination. Nevertheless, the final model still missed approximately 40.4% of melanoma cases, which means it is not suitable for independent clinical diagnosis. The results should therefore be interpreted as evidence of experimental potential, not clinical readiness.

The experiments demonstrate that evaluating medical AI systems requires more than just overall accuracy. Metrics such as melanoma recall, melanoma specificity, balanced accuracy, macro F1-score, and confusion matrices provide a more meaningful understanding of model behavior in clinically relevant situations.

6.3 Ethical Considerations

Although a publicly available, anonymized dataset is used, ethical considerations remain significant given the medical diagnostic context. The models developed are experimental and not intended for clinical application. Consequently, their results should be interpreted strictly within the scope of academic evaluation.

A primary ethical concern is the risk of false negatives, particularly regarding melanoma. Failure to identify melanoma cases may lead to a false sense of security in clinical practice. Therefore, melanoma sensitivity is prioritized as a key evaluation metric. Conversely, false positives are also significant, as misclassifying benign lesions as malignant can lead to unnecessary anxiety and additional medical procedures.

Dataset bias represents an additional ethical consideration. Public dermatology datasets often lack equal representation of all skin types, which may reduce model reliability for underrepresented populations. This limitation constrains the generalizability of findings and underscores the necessity for caution when interpreting model performance.

This thesis does not propose AI as a substitute for dermatologists. Rather, AI-based image classification is considered a potential decision-support or screening tool. Actual clinical implementation would necessitate further validation, regulatory approval, explainability, and assessment with diverse patient populations.

6.4 Interpretability and Clinical Trust

Although this thesis focuses primarily on classification performance, interpretability remains a key limitation of the evaluated models. ResNet and EfficientNet can produce class predictions, but the experiments in this thesis do not directly explain which visual features influenced each prediction. This means that even when a model correctly classifies a lesion, the reasoning behind the prediction remains difficult to verify.

This is particularly relevant in medical contexts, where clinicians need to understand whether a model is focusing on medically meaningful lesion characteristics or on irrelevant image artifacts. For example, a model might learn patterns related to lighting, image quality, or dataset-specific acquisition conditions rather than clinically relevant features such as border irregularity, color variation, or lesion structure.

Therefore, the results of this thesis should be interpreted as a performance evaluation rather than a clinically explainable diagnostic system. The lack of interpretability limits the practical use of the models in real medical workflows, even when classification metrics appear promising. Future work should include explainable AI techniques such as Grad-CAM or saliency maps to visualize which image regions influence model predictions and to evaluate whether the models rely on clinically meaningful features.

6.5 Summary of Findings

This thesis investigated the application of deep learning-based image recognition models to skin cancer classification using the HAM10000 dataset. Multiple experimental configurations were tested to evaluate how transfer learning, class imbalance handling, fine-tuning strategies, and dataset composition affect model performance.

The experiments demonstrated that overall accuracy alone is insufficient as a metric for evaluating medical classification systems. Early baseline models achieved deceptively high accuracy while failing to correctly classify minority classes such as melanoma. This highlighted the importance of using additional metrics, including sensitivity, specificity, balanced accuracy, and confusion matrices.

The study further showed that class imbalance strongly affects model behavior. Minority classes with very limited representation caused unstable training and poor generalization. Although class weighting partially improved attention to the minority class, it also introduced instability and inconsistent validation performance across several experiments.

The strongest final configuration was EfficientNetB0 with a capped class weight 3.0. It achieved the highest overall accuracy, balanced accuracy, macro F1-score, weighted F1-score, and melanoma recall among the four final configurations. Specifically, it achieved a test accuracy of 0.649, a balanced accuracy of 0.518, a macro F1-score of 0.483, a weighted F1-score of 0.683, a melanoma recall of 0.596, and a melanoma specificity of 0.800. Although ResNet50 2.0 achieved the highest melanoma specificity at 0.902, its melanoma recall was substantially lower at 0.332. Therefore, EfficientNetB0 3.0 was selected as the most suitable final model for the screening-oriented objective of this thesis.

Despite the stronger performance of EfficientNetB0 3.0, the results remain below dermatologist-level diagnostic performance reported in existing literature. The experiments, therefore, confirm both the potential and the limitations of current AI-based skin lesion classification systems. While such models may eventually support medical screening and assistive diagnostic workflows, they are not yet reliable enough to replace professional clinical evaluation.

Overall, this thesis's findings demonstrate that AI-based skin cancer diagnosis is highly dependent on dataset quality, class distribution, training methodology, and evaluation strategy. The research also highlights the importance of interpretability, balanced evaluation metrics, and realistic expectations when discussing the integration of AI systems into medical practice.

7. Conclusion

This thesis explored the use of deep learning-based image recognition models for skin cancer classification using the publicly available HAM10000 dataset. The research focused on evaluating the accuracy of convolutional neural networks in classifying dermoscopic images and how different training strategies affect diagnostic performance.

The experiments demonstrated that transfer learning architectures, such as ResNet50, can achieve meaningful classification performance, particularly when combined with fine-tuning, data augmentation, and techniques for handling class imbalance. However, the study also revealed several important limitations. Strong dataset imbalance significantly influenced predictive performance, often leading the model to favor dominant classes while struggling to correctly classify underrepresented lesion categories.

The research further showed that overall accuracy alone can be misleading in medical AI systems. Early experiments achieved relatively high accuracy but failed to reliably detect melanoma. More informative evaluation metrics, such as sensitivity, specificity, balanced accuracy, and confusion matrices, proved essential for understanding the models' true diagnostic performance.

The strongest final configuration was EfficientNetB0 with a capped class weight of 3.0. It achieved a test accuracy of 64.9%, a balanced accuracy of 51.8%, a macro F1-score of 48.3%, a weighted F1-score of 68.3%, a melanoma recall of 59.6%, and a melanoma specificity of 80.0%. These results show that EfficientNetB0 outperformed ResNet50 on the reduced five-class HAM10000 task, particularly on class-balanced and melanoma-focused metrics. Although this represents a substantial improvement over earlier experiments, the results remain below the diagnostic performance ranges reported for experienced dermatologists in the existing literature. In particular, the final model still missed approximately 40.4% of melanoma cases, which means that it should not be interpreted as suitable for independent clinical diagnosis.

The findings suggest that AI-based image classification systems may eventually provide useful support for medical screening and assistive diagnostic workflows, particularly in environments with limited access to dermatological expertise. At the same time, the experiments confirm that current systems remain highly dependent on dataset quality, balanced representation, and careful evaluation methodology. For these reasons, AI should currently be viewed as a supportive technology rather than a replacement for professional medical diagnosis.

Future work could explore larger, more diverse datasets; additional architectures or EfficientNet variants; explainable AI techniques; threshold tuning; and multimodal approaches that combine clinical metadata with image analysis.

7.1 Future Work

Future work should extend the architecture comparison by testing architecture-specific optimization strategies rather than relying solely on the same capped class-weight settings across models. Although this thesis compared ResNet50 and EfficientNetB0 under capped class weights of 2.0 and 3.0, further experiments could tune class weights, learning rates, augmentation strength, and decision thresholds separately for each architecture. This may improve the sensitivity-specificity trade-off, especially for melanoma detection.

Another potential improvement involves using larger, more diverse dermatology datasets. The HAM10000 dataset exhibits strong class imbalance and limited representation of darker skin tones, limiting models' ability to generalize reliably across broader populations. Future studies could therefore investigate multi-dataset training strategies and more balanced image collections.

In particular, future experiments could apply Grad-CAM or similar visualization methods to compare whether ResNet and EfficientNet focus on clinically relevant lesion regions or on irrelevant image artifacts.

Additional experiments could explore alternative imbalance-handling methods, such as focal loss, oversampling, synthetic image generation, or threshold tuning, for melanoma detection. Explainable AI techniques may further improve clinical trust by helping dermatologists understand which image features influence model predictions.

Finally, future systems may benefit from combining dermoscopic image analysis with additional clinical information such as patient history, lesion evolution, age, and localization. Multimodal diagnostic approaches could potentially improve both sensitivity and specificity beyond what is achievable using image classification alone.

8. Deliverables

Deliverable/Milestone	Description / Status
Dataset preparation and preprocessing	This step includes loading the HAM10000 dataset, inspecting its structure, cleaning and organizing the images, assigning class labels, and preparing the data for model training. Preprocessing tasks include resizing images, normalizing pixel values, applying data augmentation, and splitting the dataset into training, validation, and test sets. Class imbalance is also addressed using techniques such as class weighting or oversampling.
Model selection and setup	ResNet50 and EfficientNetB0 were implemented and compared under two capped class-weight configurations, 2.0 and 3.0.
Thesis document	The document presents both the technical work and its scientific relevance
Thesis Artefact	The thesis artifact consists of the experimental notebooks, model training code, saved evaluation outputs, visualizations, and supporting scripts. These files are provided in the GitLab repository: [https://github.com/RudLala/Bachelor_Thesis_AI_Skin_Cancer].

Citations

Brancaccio, G., et al. (2024). *Artificial intelligence in skin cancer diagnosis: A reality check*. Journal of Investigative Dermatology.

Chanda, R., et al. (2024). *Explainable AI for melanoma diagnosis aligned with clinical reasoning*. Nature Communications.

Daneshjou, R., et al. (2022). *Disparities in dermatology AI performance on diverse clinical images*. Science Advances.

Esteva, A., et al. (2017). *Dermatologist-level classification of skin cancer with deep neural networks*. Nature.

Haenssle, H. et al. (2018). *Man against machine: Diagnostic performance of CNN compared to dermatologists*. Annals of Oncology.

Kittler, H., et al. (2002). *Diagnostic accuracy of dermoscopy*. Lancet Oncology.

Tschandl, P., et al. (2018). *HAM10000 dataset: A large collection of multi-source dermoscopic images*. Scientific Data.

Tschandl, P., et al. (2019). *Human-computer collaboration for skin cancer recognition*. Nature Medicine.

Vestergaard, M., et al. (2008). *Dermoscopy vs naked eye for melanoma diagnosis*. British Journal of Dermatology.

He, K., Zhang, X., Ren, S., & Sun, J. (2015). Deep residual learning for image recognition. arXiv. <https://arxiv.org/abs/1512.03385>

Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. arXiv. <https://arxiv.org/abs/1905.11946>

OpenAI. (2026). ChatGPT (GPT-5.5 Thinking) [Large language model]. <https://chatgpt.com>

Grammarly, Inc. (2026). Grammarly [AI writing assistant]. <https://www.grammarly.com>

Rud, L. (2026). Bachelor_Thesis_AI_Skin_Cancer [GitHub repository]. GitHub.

https://github.com/RudLala/Bachelor_Thesis_AI_Skin_Cancer

Appendix A: Training Curves

Figure A1. Training and validation accuracy for ResNet50 2.0.

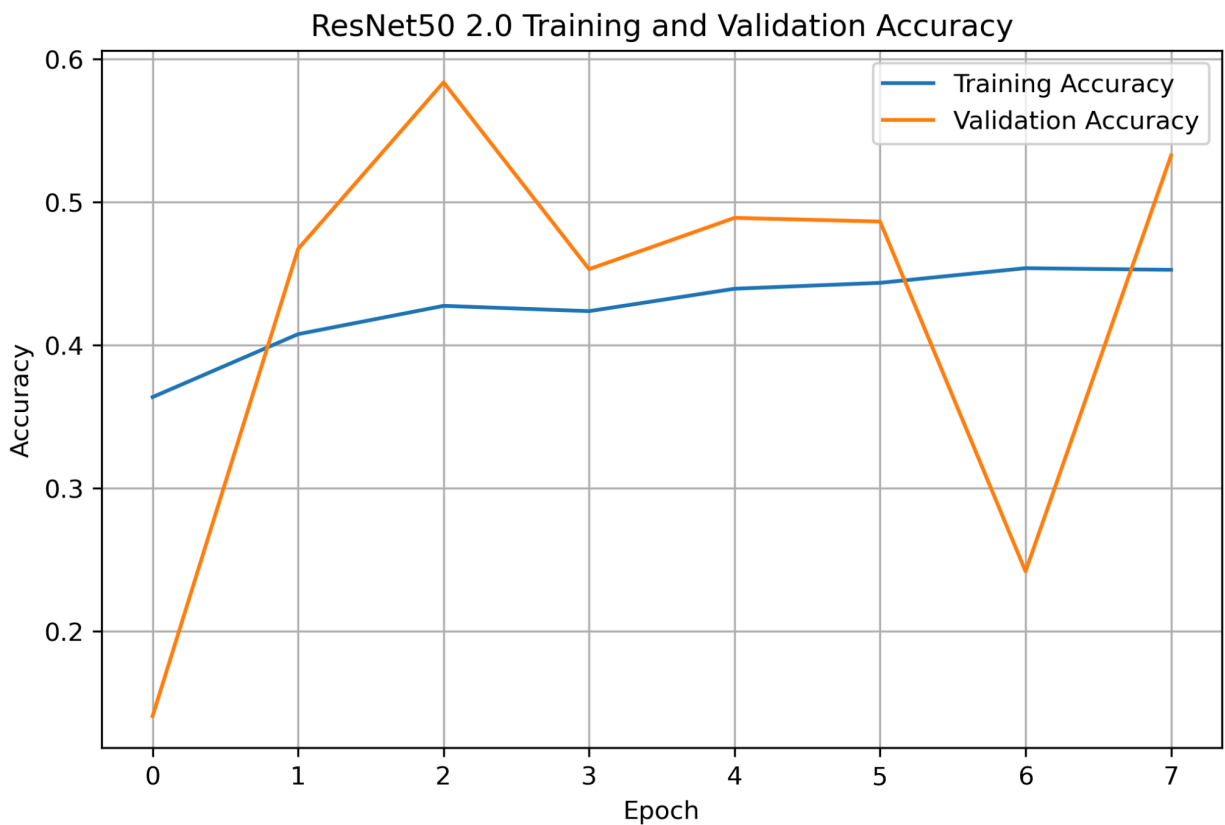


Figure A2. Training and validation loss for ResNet50 2.0.

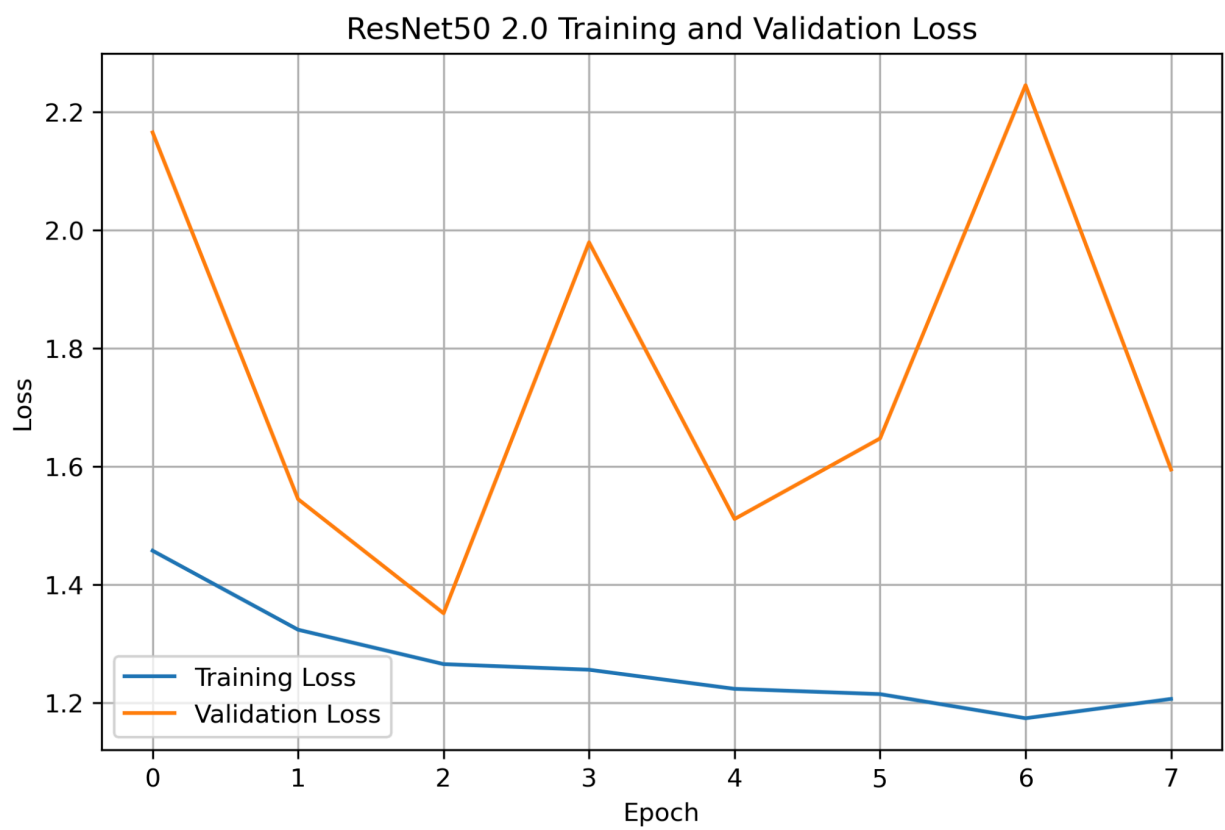


Figure A3. Training and validation accuracy for ResNet50 3.0.

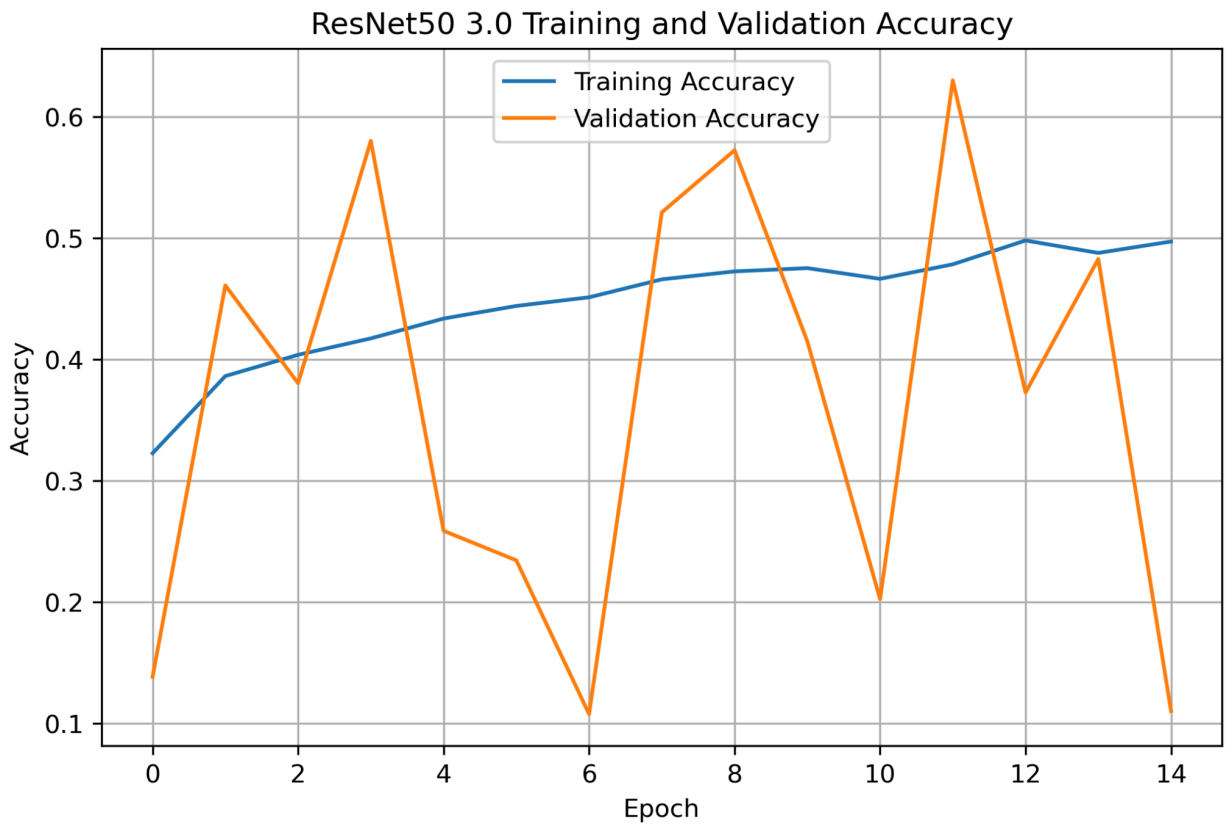


Figure A4. Training and validation loss for ResNet50 3.0.

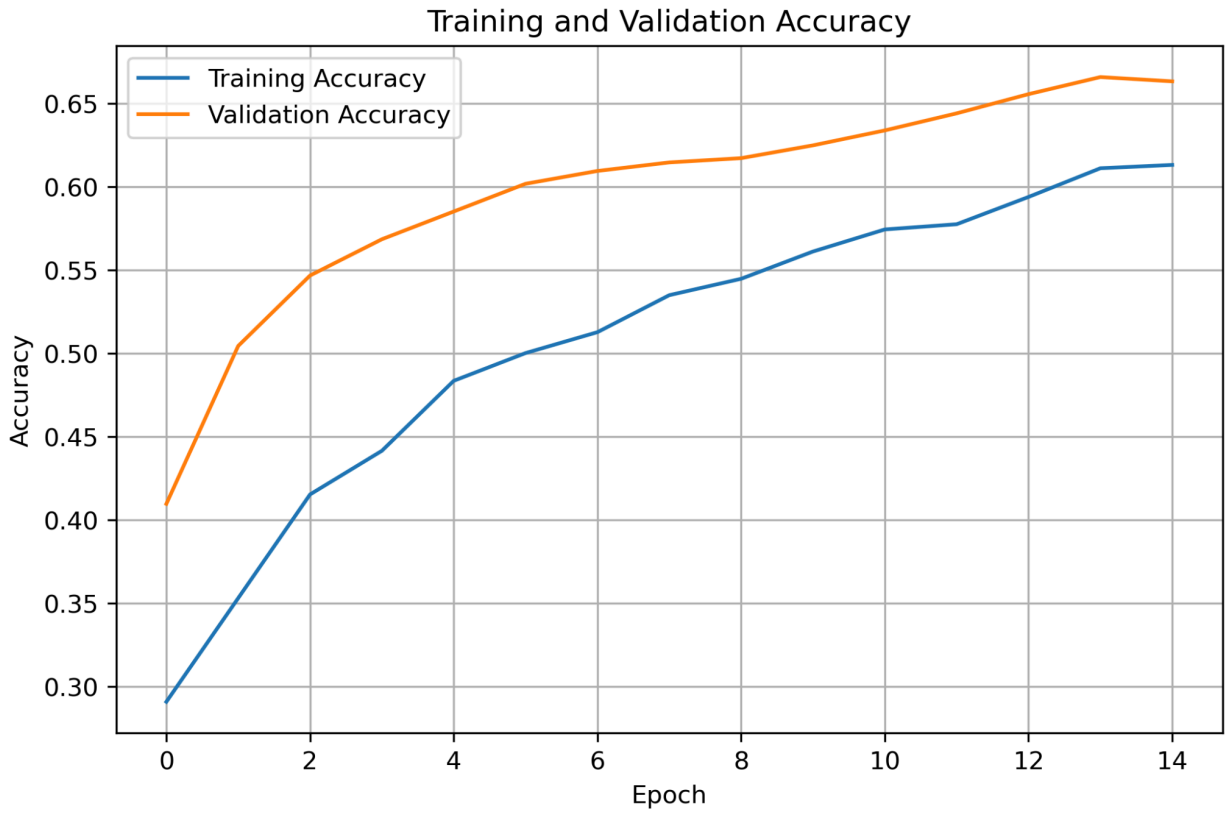


Figure A5. Training and validation accuracy for EfficientNetB0 2.0.

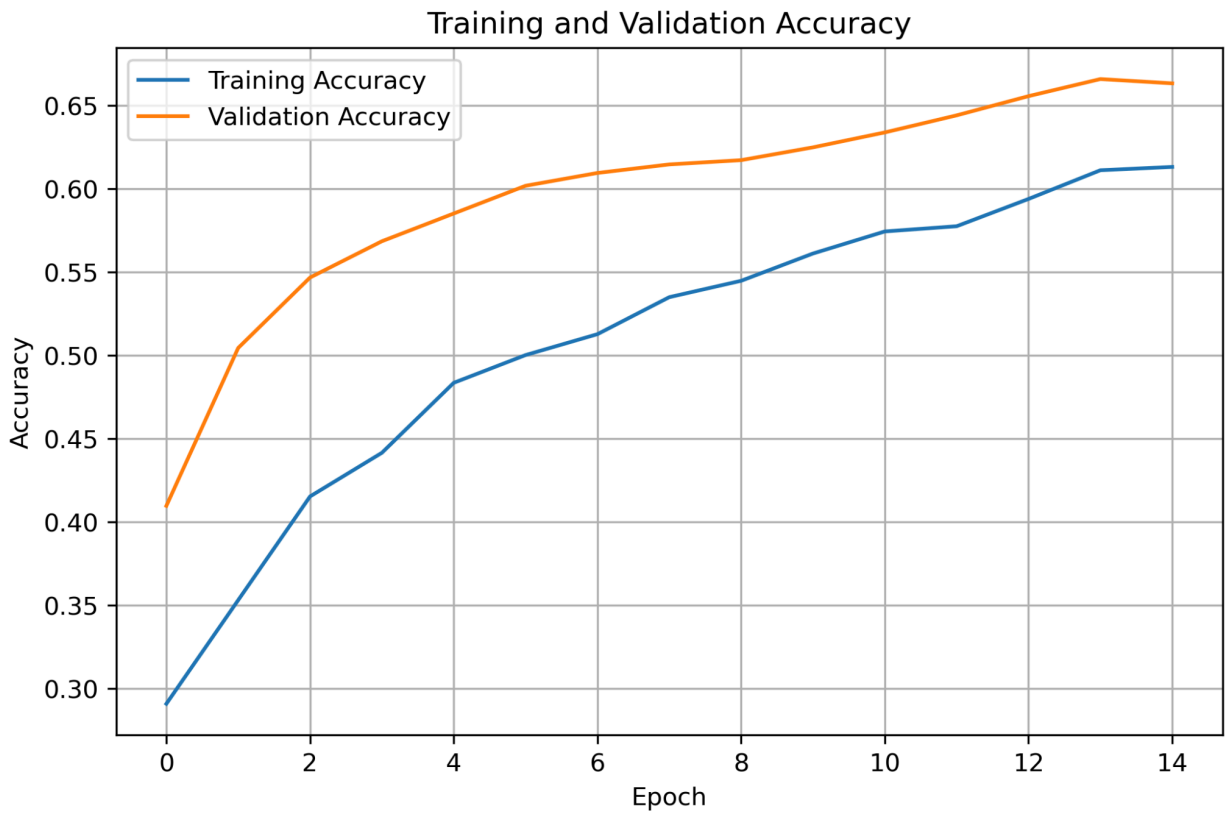


Figure A6. Training and validation loss for EfficientNetB0 2.0.

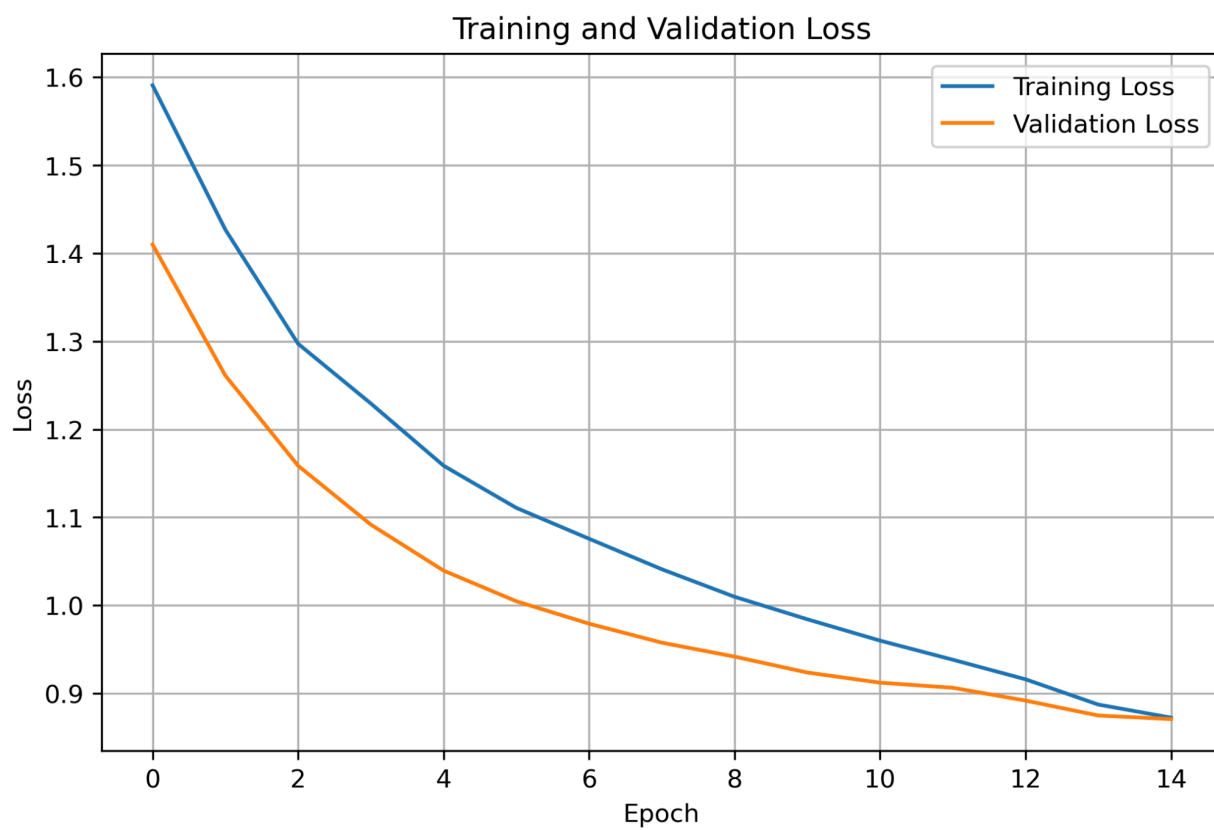


Figure A7. Training and validation accuracy for EfficientNetB0 3.0.

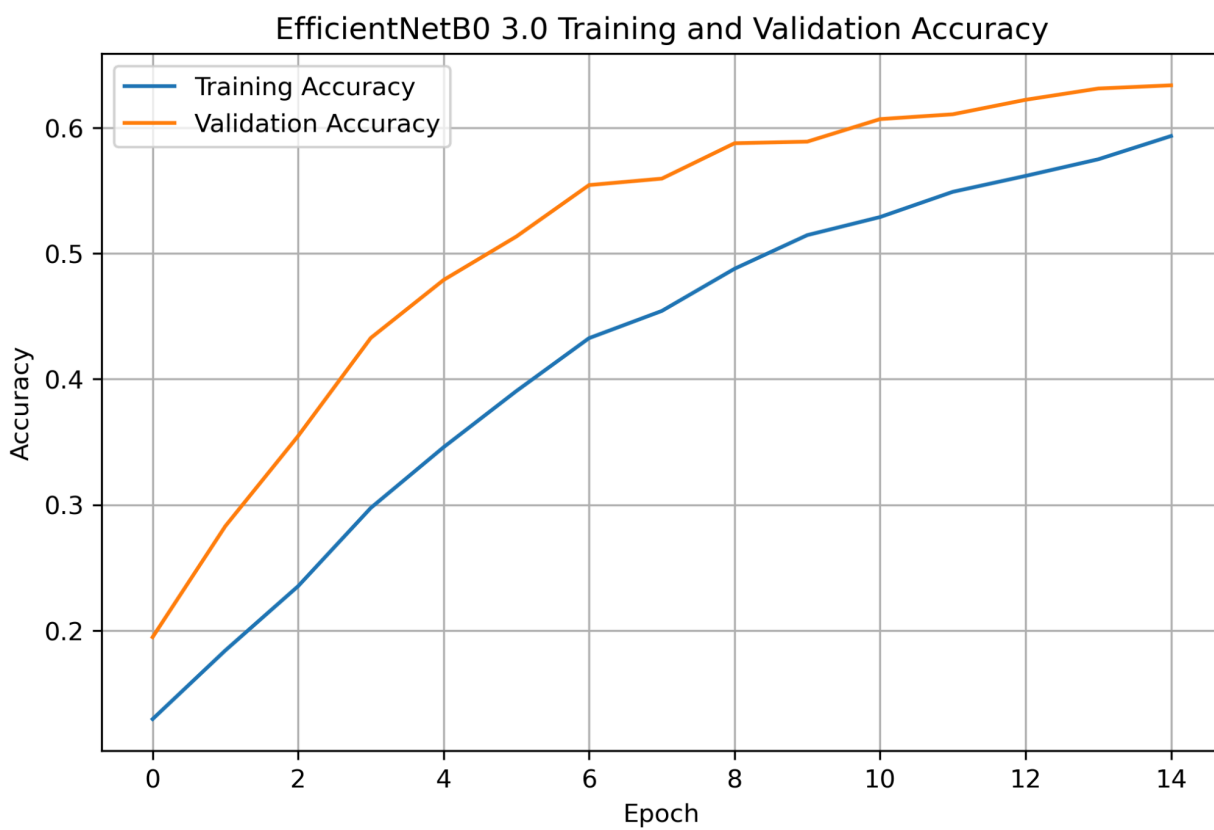
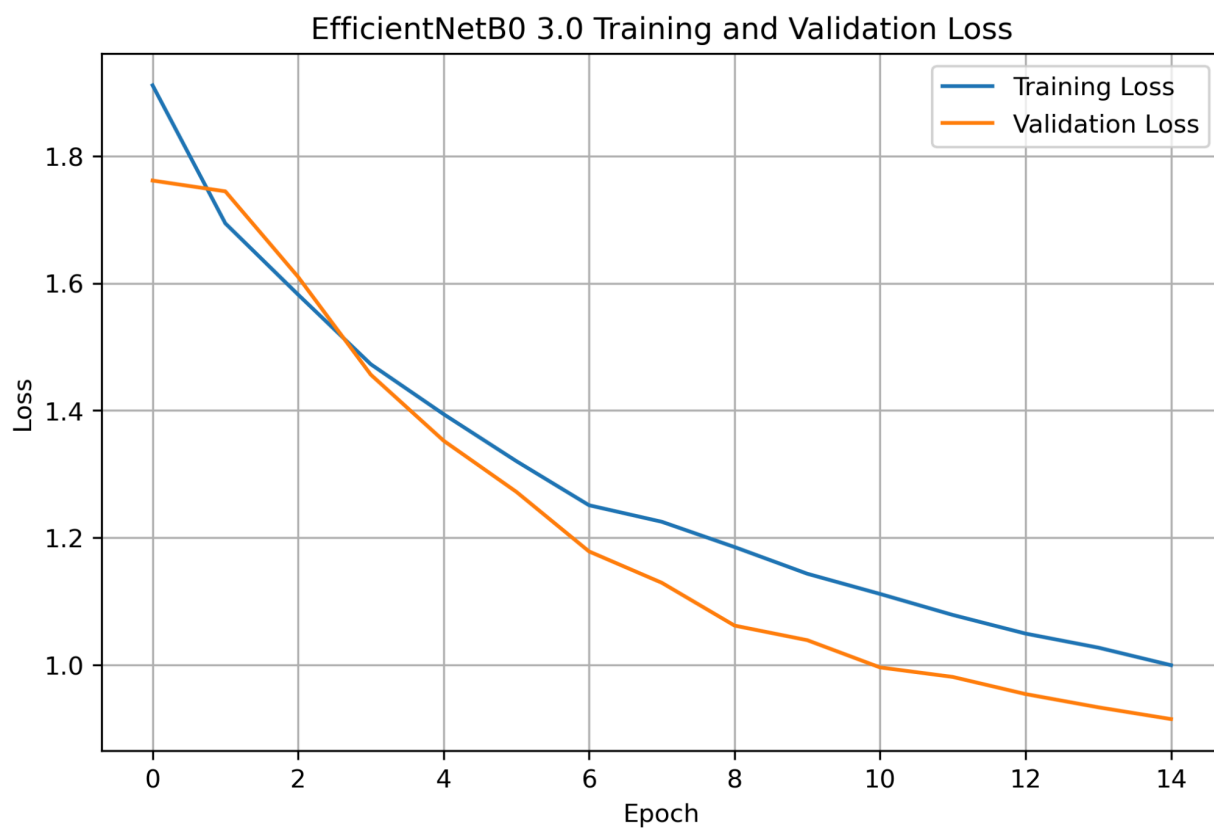


Figure A8. Training and validation loss for EfficientNetB0 3.0.



Appendix B: Confusion Matrices

Figure B1. Confusion matrix for ResNet50 2.0.

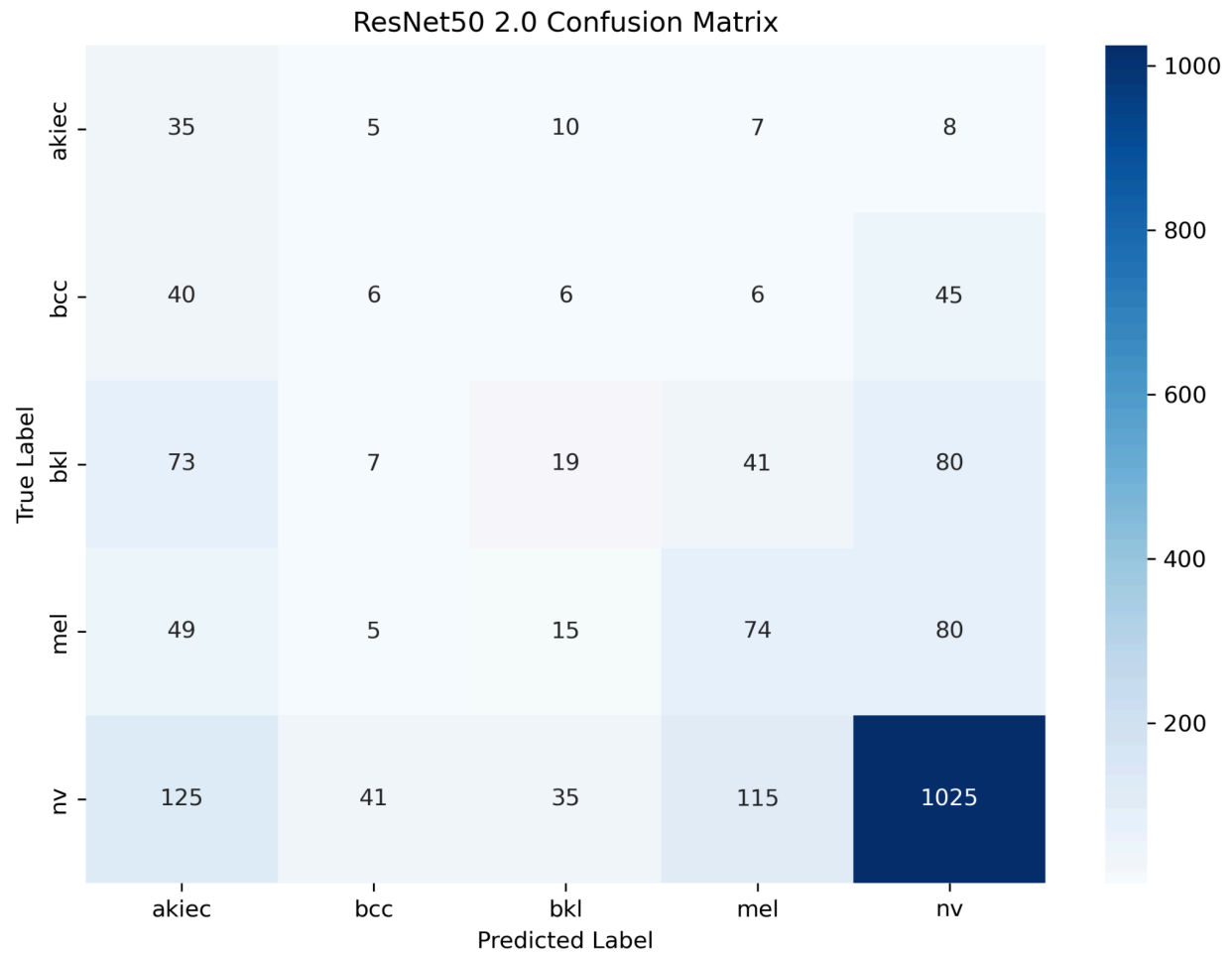


Figure B2. Confusion matrix for ResNet50 3.0.

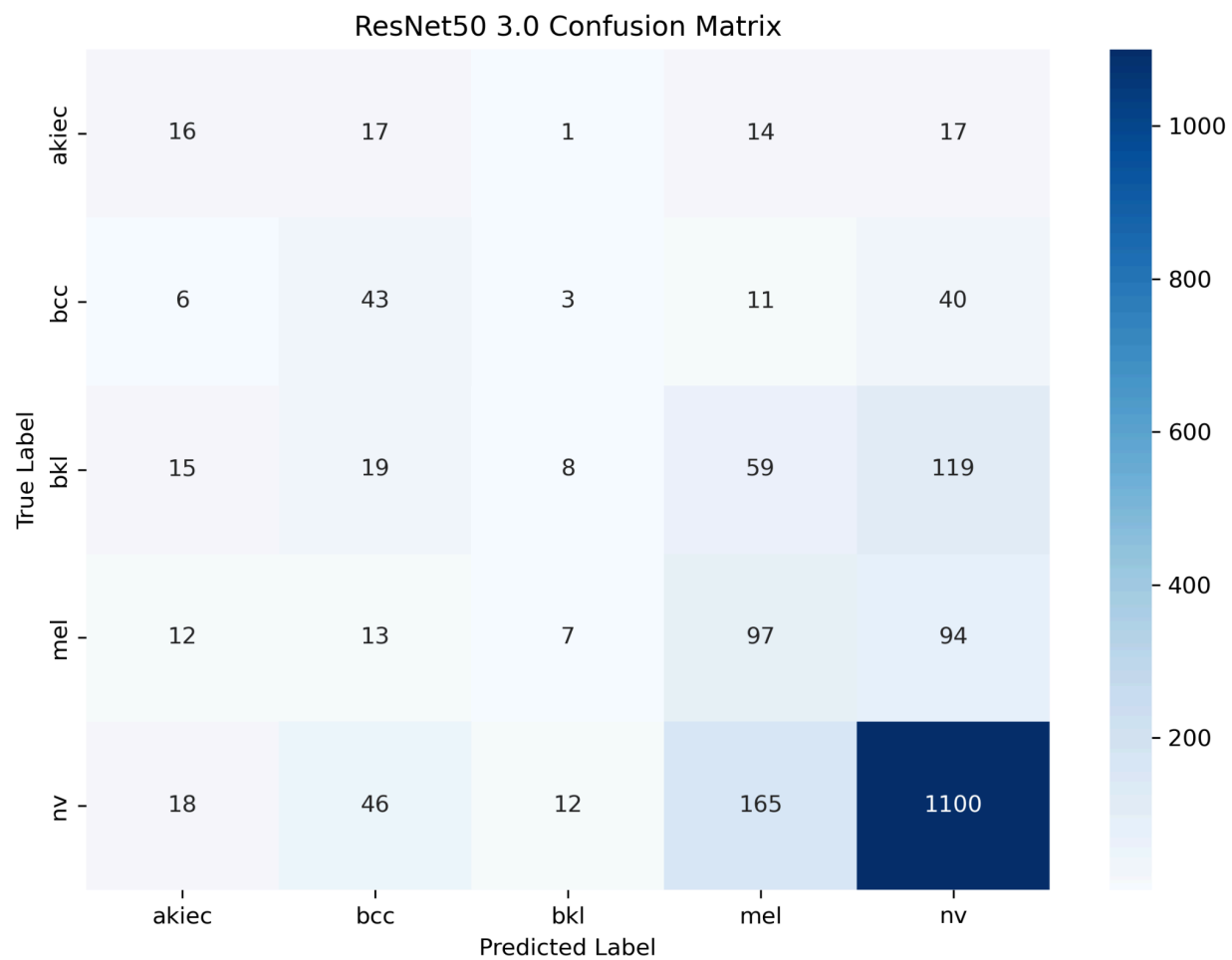


Figure B3. Confusion matrix for EfficientNetB0 2.0.

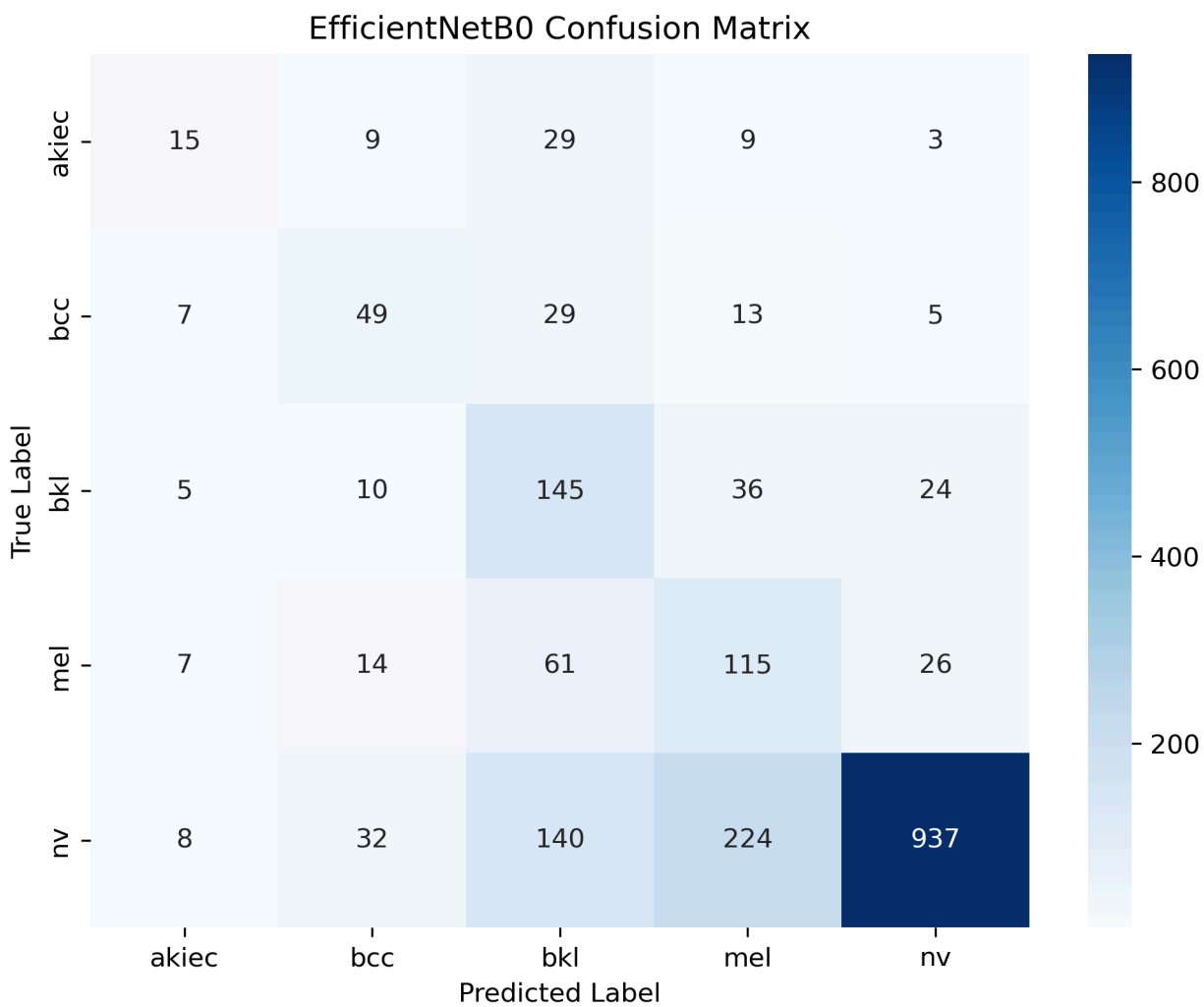


Figure B4. Confusion matrix for EfficientNetB0 3.0.

